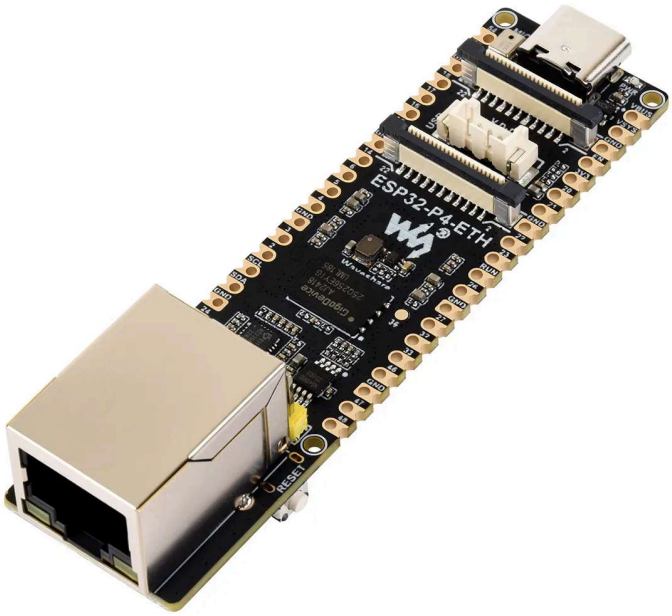
ESP32-P4-ETH



The **ESP32-P4-ETH** is a multimedia development board based on the ESP32-P4. It provides a rich set of human-machine interfaces, including MIPI-CSI (integrated image signal processor ISP), MIPI-DSI, SPI, I2S, I2C, LED PWM, MCPWM, RMT, ADC, UART, TWAI, etc. Additionally, it supports USB OTG 2.0 HS and Ethernet, and features an onboard 40PIN GPIO expansion header compatible with some Raspberry Pi Pico expansion boards for broader application adaptability. The ESP32-P4 is powered by a RISC-V dual-core processor, supports up to 32MB PSRAM, and integrates peripherals such as USB 2.0, MIPI-CSI/DSI, and an H.264 encoder, meeting the demands of low-cost, high-performance, and low-power multimedia development. In addition, the ESP32-P4 integrates a digital signature peripheral and a dedicated key management unit to ensure data and operation security. The ESP32-P4-ETH is specially designed for high-performance and high-security applications, meeting the needs of embedded systems in human-machine interaction, edge computing, and IO expansion.

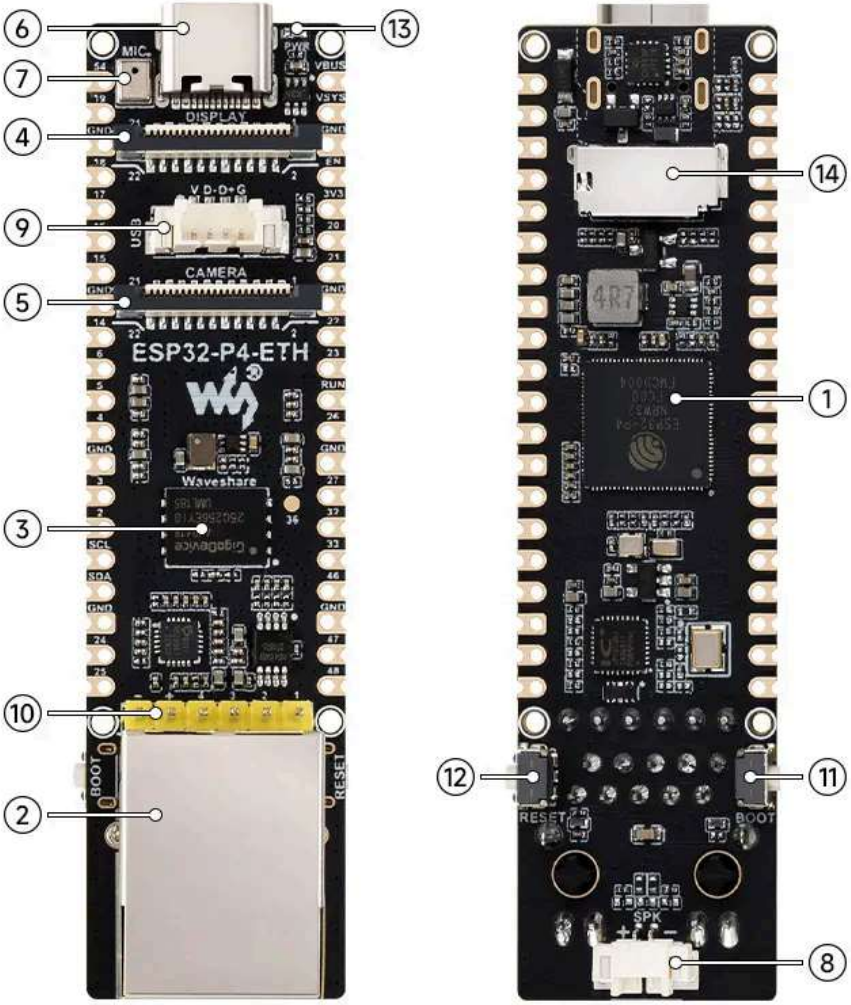
SKU	Product
32086	ESP32-P4-ETH
32087	ESP32-P4-ETH-M
34247	ESP32-P4-POE-ETH-NH
32088	ESP32-P4-POE-ETH
34248	ESP32-P4-POE-ETH-NH-KIT-A
32089	ESP32-P4-POE-ETH-KIT-A
34249	ESP32-P4-POE-ETH-NH-KIT-B
32090	ESP32-P4-POE-ETH-KIT-B

Features

- Processor
  - Equipped with a RISC-V 32-bit dual-core processor (HP system), featuring DSP and ISA extensions, a Floating-Point Unit (FPU), with a main frequency of up to 360 MHz.
  - Equipped with a RISC-V 32-bit single-core processor (LP system), with a main frequency of up to 40 MHz.
- Memory
  - 128 KB of high-performance (HP) system read-only memory (ROM).
  - 16 KB of low-power (LP) system read-only memory (ROM).
  - 768 KB of high-performance (HP) L2 memory (L2MEM).
  - 32 KB of low-power (LP) SRAM.
  - 8 KB of system tightly coupled memory (TCM).
  - 32 MB PSRAM stacked in the package, onboard 32MB Nor Flash
- Peripheral Interfaces
  - 2 × 20 pin headers onboard, exposing 27 remaining programmable GPIOs
  - Onboard speaker interface and microphone, enabling ideal audio functionality using the Codec chip and amplifier chip

- Onboard MIPI-CSI high-definition camera interface, supporting Full HD 1080P image capture and encoding, integrated Image Signal Processor (ISP), H.264 video encoder, supports H.264 & JPEG video encoding (1080P @30fps), facilitating applications in computer vision, machine vision, and other fields
- Onboard MIPI-DSI high-definition display interface, integrated Pixel Processing Accelerator (PPA), 2D graphics acceleration controller (2D DMA), supports JPEG image decoding (1080P @30fps), providing strong support for high-definition display and smooth HMI experiences, suitable for applications such as smart home control panels, industrial control panels, and vending machines

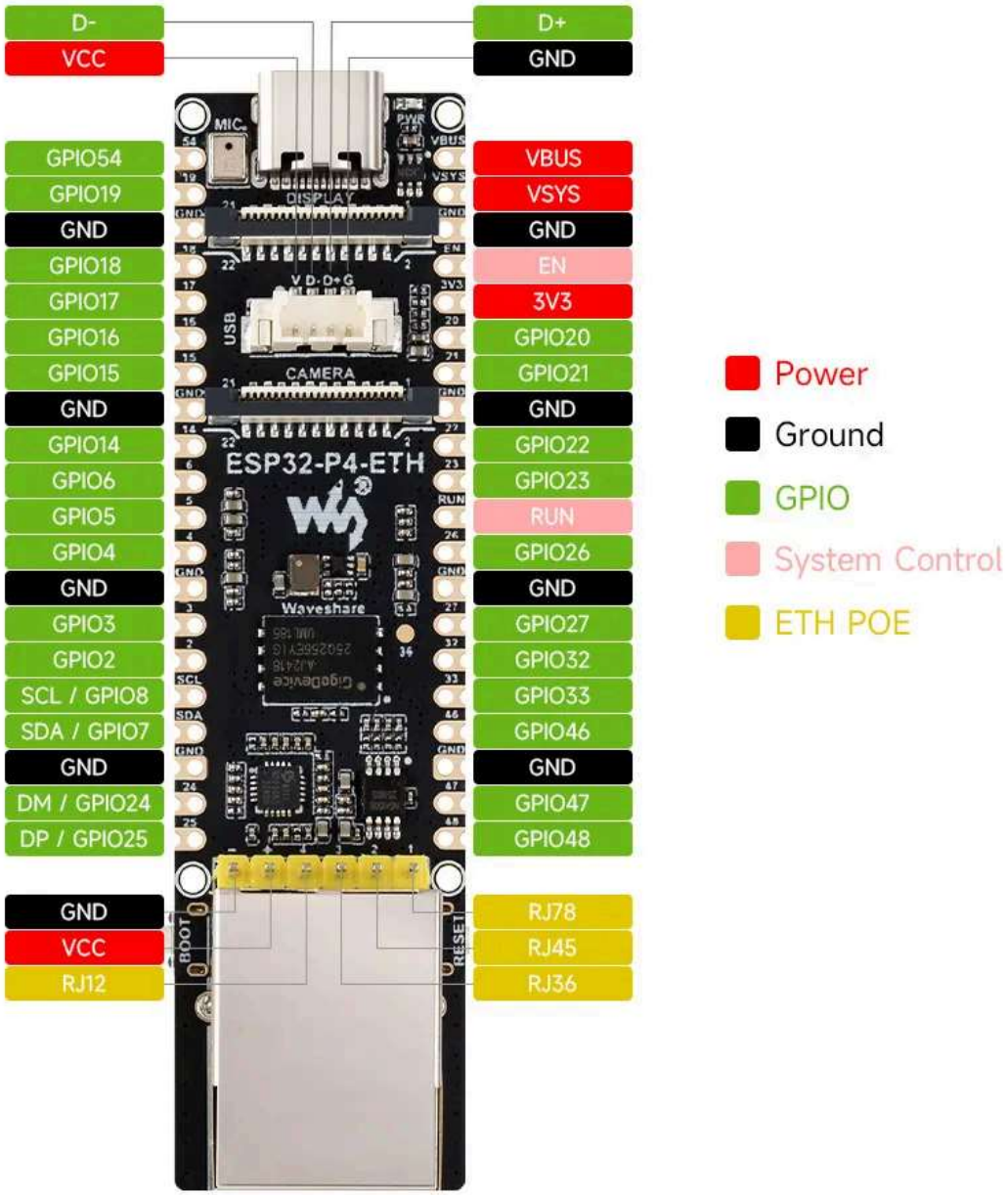
Hardware Description



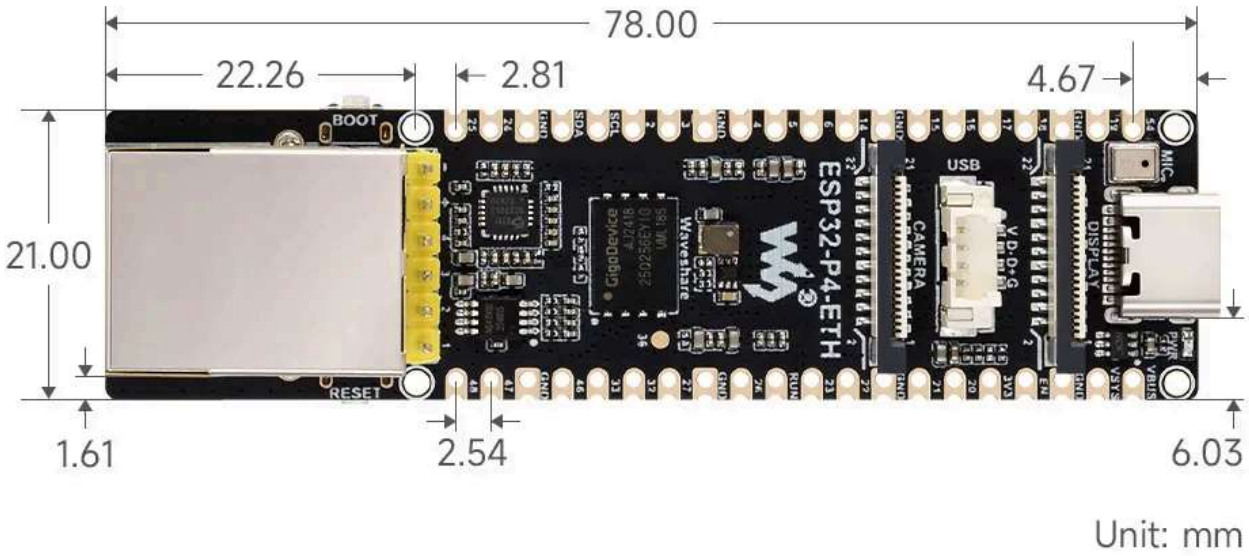
1. **ESP32-P4NRW32** ESP32-P4 with 32MB stacked PSRAM
2. **100M RJ45 Ethernet Port**
3. **32MB Nor Flash**

4. **Display Interface** MIPI-DSI (2-lane), compatible with `5` / `7` / `8` / `10.1` inch DSI screens
5. **Camera Interface** MIPI-CSI (2-lane), compatible with cameras such as `OV5647`
6. **Type-C Interface** For power supply, programming, and debugging
7. **SMD Microphone**
8. **Speaker Interface** MX1.25 2P connector, supports 8Ω 2W speaker
9. **4PIN USB Interface** USB OTG 2.0 High Speed interface
10. **PoE Module & Power Supply Header**
11. **BOOT Button** Press during power-up or reset to enter download mode
12. **RESET Button** Reset button
13. **Power Indicator**
14. **TF Card Slot** SDIO 3.0 interface protocol

Pinout Definition


D-
VCC
D+
GND
GPIO54
GPIO19
GND
GPIO18
GPIO17
GPIO16
GPIO15
GND
GPIO14
GPIO6
GPIO5
GPIO4
GND
GPIO3
GPIO2
SCL / GPIO8
SDA / GPIO7
GND
DM / GPIO24
DP / GPIO25
VBUS
VSYS
GND
EN
3V3
GPIO20
GPIO21
GND
GPIO22
GPIO23
RUN
GPIO26
GND
GPIO27
GPIO32
GPIO33
GPIO46
GND
GPIO47
GPIO48
GND
VCC
RJ12
RJ78
RJ45
RJ36
MIC
DISPLAY
USB
CAMERA
ESP32-P4-ETH
Waveshare
BOOT
RESET
Power
Ground
GPIO
System Control
ETH POE

Dimensions



Development Tools

Each of these two development approaches has its own advantages, and developers can choose according to their needs and skill levels. Arduino is suitable for beginners and non-professionals because they are easy to learn and quick to get started. ESP-IDF is a better choice for developers with a professional background or high performance requirements, as it provides more advanced development tools and greater control capabilities for the development of complex projects.

> **WARNING**
>
> ESP32-P4 currently has limited adaptation on the Arduino platform. To ensure development stability, it is recommended to use ESP-IDF for development at this stage. If you have requirements for using the Arduino platform, you can participate in the discussion on GitHub: Support of Arduino

- **ESP-IDF**, or full name Espressif IDE, is a professional development framework introduced by Espressif Technology for the ESP series chips. It is developed using the C language, including a compiler, debugger, and flashing tool, etc., and can be developed via the command lines or through an integrated development environment (such as Visual Studio Code with the Espressif IDF plugin). The plugin offers features such as code navigation, project management, and debugging, etc. We recommend using VS Code for development. For the specific configuration process, please refer to the **Working with ESP-IDF**. The tutorial also provides relevant demos for reference.

Give Feedback